When Neural Collapse Fails to Route: Routing Drift in Continual Learning

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Abstract

Progressive neural-collapse classifiers can learn strong task-local decision boundaries while failing the global Class-IL decision. We identify the failure as *routing drift*: under an expanding equiangular tight frame (ETF) classifier, a held-out sample can remain separable within its true task block, yet receive its largest global score from another block. This produces high Task-IL accuracy but low Class-IL accuracy, and small replay buffers can mask the problem because stored samples are optimized repeatedly and often remain well routed. We formalize route accuracy directly from the global class argmax and use it to separate within-task discrimination from cross-task score comparability. This decomposition suggests a classifier-level correction: preserve the global ETF frame, but add class-local reliability evidence when transported class measures deviate from the neural-collapse ideal. We develop Reliability-Calibrated Routing (RCR), a lightweight post-hoc classifier that composes standardized ETF alignment with radial plausibility around transported, NC-shrunk class representatives. Across three datasets and two fixed replay-buffer budgets, this NC-consistent correction gives the best Class-IL FAA on five of six settings and reduces Class-IL forgetting on all six. The largest gains occur in harder routing regimes: up to +6.75 FAA over the reported reference, and up to +8.03 FAA and +8.62 final route accuracy in paired evaluations, while preserving Task-IL performance. These results position routing drift as a measurable failure mode of ETF-based continual learning and show that reliability-calibrated class evidence can repair part of the lost global routing without retraining the representation. Code and evaluation artifacts are available at this URL.

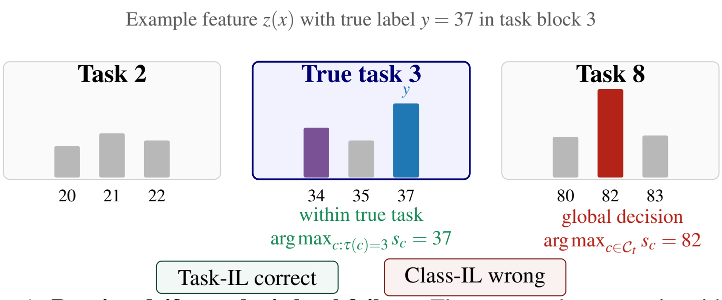


Figure 1: **Routing drift as a logit-level failure.** The correct class can win within its task block while a different block contains the global argmax. Routing drift is therefore not a within-task class error; it is a cross-task score-comparability error.

1 Introduction

Class-incremental learning (Class-IL) asks a model to classify among all classes observed so far, without receiving the task identity at test time. Task-incremental learning (Task-IL) is easier: the correct task block is known, so the classifier only chooses among classes inside that block. The gap between Class-IL and Task-IL therefore measures more than forgetting; it reveals whether evidence remains comparable across task blocks [3, 25, 28]. Prior analyses decompose Class-IL into selecting the correct task and selecting the correct class within that task [16, 17, 22]. We study the score-level form of this decomposition inside progressive ETF classifiers.

Neural-collapse-inspired continual learners use equiangular tight frame (ETF) classifiers to impose a structured global geometry on class prototypes [27]. ProNC is a representative example: it expands the ETF as new classes arrive and learns features aligned to the evolving frame [36]. This geometry yields strong Task-IL performance, which means task-local class separation is often preserved. Yet our diagnostics show that many remaining Class-IL errors are not same-task confusions. The correct class can be identifiable once the true task block is known, while the global ETF score assigns the sample to another block.

Figure 1 illustrates this score-level failure. The true class has the largest logit inside its task block, so the Task-IL decision is correct, but another block contains the global maximum. We call this *routing drift*. In ETF continual-learning runs, Task-IL can remain high while route accuracy collapses, indicating that the model keeps task-local discrimination but loses global route comparability.

This paper studies routing drift as a geometric failure mode of neural-collapse continual learning. The point is not that ETF classifiers fail to separate classes locally; rather, their global scores become unreliable across task blocks. We therefore ask whether Class-IL can be improved by correcting cross-task competition while preserving the task-local structure already learned by the progressive ETF model. Our answer is Reliability-Calibrated Routing (RCR), a reliability-compositional score that combines global ETF alignment with radial plausibility around transported class measures. RCR targets the inference geometry of progressive ETF classifiers: it keeps the global neural-collapse frame, adds local class reliability, and does not use an external task predictor. It is a post-hoc classifier-level correction, not a new representation learner; this scope is deliberate because it isolates the routing failure from the underlying training dynamics. The method is designed for progressive ETF classifiers, and extending the same reliability principle beyond explicit ETF frames is left open.

Our contribution is threefold. **(i)** We define route accuracy as a diagnostic read directly from the global ETF prediction, specializing the known Class-IL decomposition to progressive neural-collapse classifiers. **(ii)** We show that replay buffers can mask routing drift, so replay-fitted task calibration is not a reliable substitute for held-out routing evaluation. **(iii)** We introduce RCR, a reliability-calibrated classifier that preserves the neural-collapse frame, models local class plausibility, gives the best Class-IL FAA on five of six dataset-buffer settings, and reduces Class-IL forgetting on all six while preserving Task-IL accuracy.

2 Related Work

Class-incremental continual learning. Class-incremental learning evaluates a model after each task over all classes seen so far, without providing the task identity at test time [24]. Early and still influential approaches regularize important parameters or logits [18, 21, 41], or rehearse previous data through direct replay [29, 30], exemplar classifiers [28], gradient constraints [1, 5, 23], and regularized replay [2, 3]. Other lines improve classifier balance or representation transfer with cosine classifiers, pooled-output distillation, feature boosting, prompts, contrastive objectives, coreset selection, and provable memory construction [4, 6, 7, 8, 15, 31, 33, 34, 37]. These methods address forgetting in the representation and in the classifier, but Class-IL also requires scores from different tasks to remain comparable. Prior analyses separate this requirement into two events: selecting the correct task block and selecting the correct class inside that block [16, 17, 22]. We use this decomposition diagnostically, but do not train a task predictor. Route accuracy is read directly from the global class argmax, and RCR changes the class score itself rather than fitting a separate task-id module.

Neural collapse and ETF classifiers. Neural collapse describes a terminal training geometry in which within-class variability shrinks, class means align with classifier weights, and these directions form a simplex ETF [27]. Subsequent work has analyzed its dynamics, layer-peeled models, imbalanced regimes, and feature-classifier alignment beyond standard balanced classification [9, 13, 38, 39]. This geometry has motivated fixed ETF classifiers for continual learning, where the classifier is prescribed rather than learned from scratch at every task [40]. Neural-collapse-inspired contrastive continual learning further uses predefined hard prototypes and simplex-structure distillation to reduce representational overlap and stabilize learning across tasks [26]. Progressive neural-collapse methods avoid committing to a final class set by expanding the ETF as new classes arrive, then using nearest-ETF inference in the current frame [36]. This design gives a simple global geometry for all classes seen so far. Our work starts from this trained progressive ETF model and asks a different question: after several feature-frame updates, is the nearest-ETF score still calibrated for both old and new classes? The answer motivates a classifier-level correction rather than a new ETF construction.

Prototype drift and local reliability. Prototype and cosine classifiers are widely used when class evidence is expressed by normalized features, class means, or learned weight vectors [35]. They are also sensitive to representation drift: an old class mean estimated in one feature frame may no longer be reliable after the backbone changes. Learnable Drift Compensation (LDC) estimates a previous-to-current feature projector and applies it to old class prototypes before nearest-class-mean inference [10]. This line of work shows that prototype positions should be interpreted in the current feature frame, not as fixed historical measurements. A complementary issue is metric reliability. FeCAM models class distributions with covariance-aware distances, showing that a single isotropic nearest-prototype rule can be too restrictive when local class geometry differs across classes [11]. We use these observations in a constrained way: transported prototypes and competitive radii provide local reliability evidence, while the explicit ETF classifier remains the global neural-collapse frame. Thus RCR is not transported NCM, not a replay-fitted calibration head, and not a replacement for ProNC's progressive ETF.

Routing terminology. In many-task learning, task routing can denote an architectural mechanism that selects task-specific paths [32]; recent mixture-of-experts continual-learning papers similarly study expert assignment drift when samples are reassigned to newly added experts [12, 42]. These works use an explicit router, expert assignment, or task-specific execution path. Our setting is different: there is one shared classifier, no test-time task identity, and no learned routing head. Routing drift denotes inter-block score incomparability inside the same global classifier. This distinction is important experimentally: improving route accuracy here means changing the class evidence used by the global argmax, not adding a task oracle in front of the classifier.

3 Preliminaries

Continual learning protocol. We consider a sequence of tasks $\mathcal{T} = \{1, \dots, T\}$. Task t provides data $\mathcal{D}_t = \{(x_i, y_i)\}$ whose labels are disjoint from previous tasks. Let $\mathcal{C}_t$ be the set of classes observed after task t, and let $\tau(c)$ map class c to its task block. In Class-IL, the model predicts among all classes in $\mathcal{C}_t$ without receiving task identity. In Task-IL, prediction is restricted to the true task block. We report final average accuracy (FAA) and final forgetting (FF). If a_i^t is the accuracy on task i after training task t, then

$$\mathrm{FAA} = \frac{1}{T}\sum_{i=1}^{T} a_i^T, \qquad \mathrm{FF} = \frac{1}{T-1}\sum_{i=1}^{T-1} \max_{1\le u<T} a_i^u - a_i^T. \tag{1}$$

Primary notation is summarized in the supplementary material (Table 2).

Route accuracy. Following the decomposition discussed in Section 2, we define a route diagnostic directly from the global class prediction. For a sample (x, y) and a global prediction $\hat{y}$,

$$\text{Class-IL}(x) = \mathbb{1}[\hat{y} = y], \tag{2}$$

$$R(x) = \mathbb{1}[\tau(\hat{y}) = \tau(y)]. \tag{3}$$

Route accuracy is the expectation of $R(x)$ over the test set. It is not an oracle available to the method, but a measurement of whether the class argmax falls inside the correct task block. High Task-IL but low route accuracy means that features remain separable within the correct block, but the global classifier assigns them to another block.

Neural collapse and ETF inference. Neural collapse describes the terminal regime in which centered class means and classifier weights align to a simplex equiangular tight frame (ETF) [27]. We recall the standard NC1–NC4 definitions in the supplementary material (Section C). A K-class simplex ETF is a set of unit vectors $\{u_c\}_{c=1}^{K}$ satisfying

$$\langle u_i, u_j \rangle = -\frac{1}{K-1} \quad \text{for } i \neq j. \tag{4}$$

Under the neural-collapse idealization, classification reduces to nearest-prototype inference in the ETF frame:

$$\hat{y}_{\mathrm{ETF}}(x) = \arg\max_{c\in\mathcal{C}_t} \langle z(x), u_c \rangle, \qquad z(x) = \frac{f_\theta(x)}{\|f_\theta(x)\|}. \tag{5}$$

Progressive Neural Collapse. ProNC progressively constructs the ETF target during continual learning rather than fixing a global ETF in advance [36]. After task 1, it extracts an ETF aligned with the learned first-task class means. Before each later task, it expands the ETF by adding vertices for new classes while keeping the old geometry close to the previous frame. For task t, the reference training objective combines supervised learning, alignment to the expanded ETF, and feature distillation from the previous model:

$$\mathcal{L}_{\text{ProNC}} = \mathcal{L}_{\text{ce}} + \lambda_1 \mathcal{L}_{\text{align}}(z, u_y) + \lambda_2 \mathcal{L}_{\text{distill}}(z, z^-), \tag{6}$$

where z^- is the normalized feature from the previous model. At evaluation, the linear classifier is replaced by nearest-ETF inference:

$$s_{\text{ETF}}(x,c) = \langle z(x), u_c \rangle, \qquad \hat{y} = \arg\max_{c \in \mathcal{C}_t} s_{\text{ETF}}(x,c). \tag{7}$$

This nearest-ETF rule is the reference classifier analyzed below. Definitions of $\mathcal{L}_{\text{align}}$ and $\mathcal{L}_{\text{distill}}$ are given in the supplementary material (Section B).

4 The Routing Drift Problem

ProNC gives every seen class an ETF vertex and compares normalized features to all vertices in one shared frame. This is the right geometry for global Class-IL: all class vertices have equal norm and equal pairwise angle, so the classifier layout itself does not favor a task block. That same symmetry makes the failure interpretable. If a wrong task block wins under an ETF head, the problem is not an uneven classifier layout. It is a mismatch between the symmetric decision rule and the class feature distributions that occupy it. Such a mismatch can emerge after several tasks because old classes are represented through replay and distillation, whereas new classes are observed densely in the current task.

To make the failure explicit, fix an evaluation stage and let b index task blocks. Write the best score inside block b as

$$m_b(x) = \max_{c \in \mathcal{C}_t : \tau(c) = b} s_{\text{ETF}}(x,c). \tag{8}$$

As summarized in Figure 2, Class-IL succeeds only if the true block wins the cross-task competition and the correct class wins inside that block. Task-IL measures only the second event. Hence a model can have high Task-IL while failing globally whenever

$$\Delta_{\text{route}}(x) = m_{\tau(y)}(x) - \max_{b \neq \tau(y)} m_b(x) < 0. \tag{9}$$

We call this *routing drift*: features remain discriminative within their own task block, but their global ETF scores no longer keep the correct block ahead. In neural-collapse terms, this is a violation of equal class-measure reliability, not a failure of the ETF geometry itself.

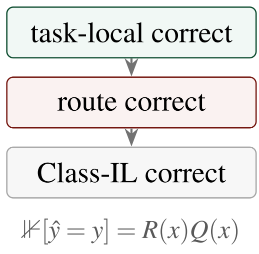


Figure 2: **Class-IL needs both events.** A prediction is globally correct only when the true task block wins and the correct class wins inside that block.

This gives a direct link between routing and Class-IL. Define the block-restricted prediction

$$\hat{y}_b(x) = \arg\max_{c \in \mathcal{C}_t : \tau(c) = b} s_{\text{ETF}}(x,c). \tag{10}$$

Let $R(x) = \mathbb{1}[\tau(\hat{y}) = \tau(y)]$ be the routing event, and let $Q(x) = \mathbb{1}[\hat{y}_{\tau(y)}(x) = y]$ be the task-local event obtained by restricting prediction to the true block. Exact Class-IL correctness requires both events:

$$\mathbb{1}[\hat{y} = y] = R(x)\,Q(x). \tag{11}$$

Thus

$$\mathrm{Acc}_{\text{Class-IL}} = \Pr(R=1)\,\Pr(Q=1 \mid R=1). \tag{12}$$

Task-IL estimates the block-local event Q, whereas Class-IL also requires the route R. When task-local discrimination is preserved, improving route accuracy directly improves achievable Class-IL. This factorization is consistent with prior analyses that decompose Class-IL into within-task prediction and task-identity prediction [16, 17, 22]. Here, route accuracy is not produced by a learned task-id predictor and is not an oracle; it is read from the global ETF classifier's own argmax.

An example block-score profile is shown in the supplementary material (Figure 3); the main diagnostic in Figure 3 shows that this is not an isolated sample-level artifact but a sequence-level failure mode.

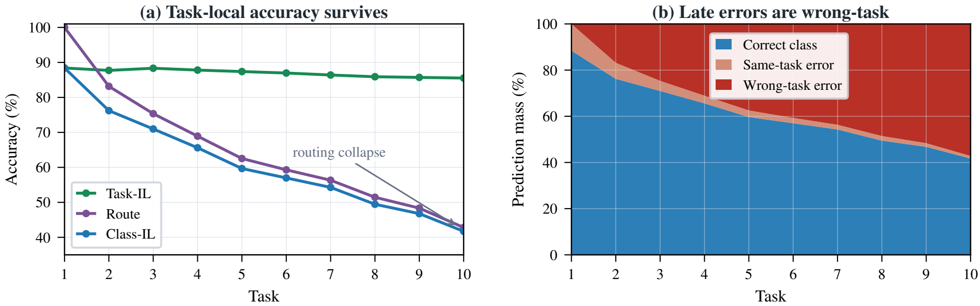


Figure 3: **Held-out errors are dominated by wrong-task routing.** On Seq-CIFAR-100 with buffer 200, Task-IL remains high across tasks, but Class-IL follows route accuracy downward. The right panel decomposes global predictions: late errors are mostly wrong-task routing rather than same-task class confusion.

Replay is also an unreliable route-calibration set. Stored samples are optimized repeatedly and can remain close to their vertices: in Figure 4, replay routing is almost perfect at the final task, while held-out test samples have only 42.81% route accuracy and 57.19% wrong-task error. A useful correction should therefore preserve the ETF frame, avoid fitting a task oracle from the buffer, and add local class reliability directly to the global scoring rule. Replay labels reveal which task a stored sample came from, but they do not measure how held-out old samples compete against newer task blocks. The classifier must instead keep the global ETF comparison intact and adjust class evidence only when the measured class

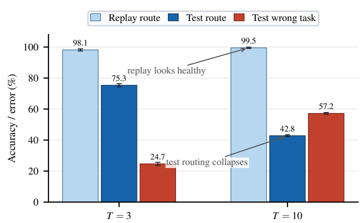
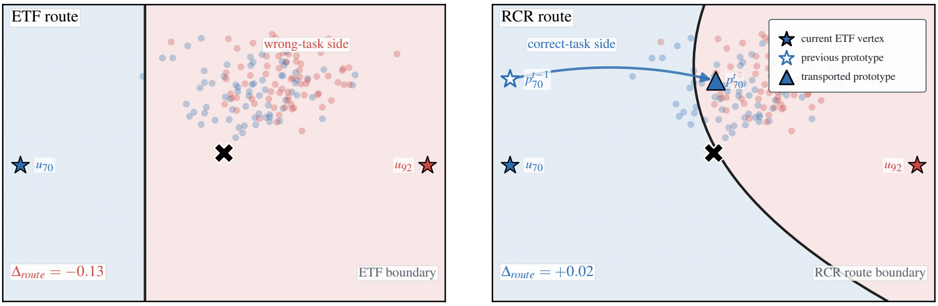


Figure 4: **Replay routing is optimistic.** Replay samples stay well routed while held-out test routing collapses on Seq-CIFAR-100/200; mean ± std over three seeds.



(a) ETF route.

(b) RCR route.

Figure 5: **RCR changes the route boundary without changing the backbone.** A held-out CIFAR-100 feature z is shown in a local two-dimensional slice passing through the original high-dimensional feature, so z has the same scores as in evaluation. The black curve is the route boundary between the correct task block and the ETF-predicted wrong block in that slice. (a) Under ETF scoring, z lies in the red region: a wrong task block wins globally even though nearby features from the correct class remain coherent. (b) RCR keeps ETF evidence but adds transported radial plausibility around p_c^t, moving the route boundary so the same feature falls in the correct-task region. Filled stars denote current ETF vertices, hollow stars denote previous prototypes, and triangles denote transported prototypes. Classes are selected only to visualize one corrected routing case; this selection is not used for training, hyperparameter choice, or evaluation.

distribution has drifted away from its ideal NC geometry. This is the motivation for RCR, which we introduce in the next section.

5 Method: Reliability-Calibrated Routing

5.1 From Routing Drift to Reliability-Calibrated Scores

Routing drift calls for a classifier-level correction. The global ETF frame should remain the shared comparison space, but each class score should also reflect how reliable its local class measure is in the current feature frame. Let $z = f_\theta(x)/\|f_\theta(x)\|$ and let u_c be the ETF vertex for class c. The reference score is

$$s_{\mathrm{ETF}}(x,c) = \langle z, u_c \rangle\,. \tag{13}$$

This score is globally comparable when all classes follow the same neural-collapse geometry, but it treats every vertex as an equally reliable point representative. RCR keeps this global NC/ETF evidence and adds local reliability around a transported prototype p_c and radius ρ_c.

5.2 Transported Class Measures

Local reliability must be measured after feature drift. Following the prototype-drift motivation of LDC [10], we estimate a previous-to-current linear transport. Unlike transported

NCM, however, we use transported prototypes only as local anchors; the ETF vertices remain the global classifier. Thus p_c is not a replacement for u_c. It is the measured location of class c in the current feature frame. Figure 5 illustrates this separation.

At the end of task t, the previous frozen backbone and the current backbone provide paired features (z_i^-, z_i) on the available training stream. We fit the same form of ridge linear transport for all classes,

$$A_t = \arg\min_A \sum_i \|z_i^- A - z_i\|_2^2 + \lambda \|A\|_F^2. \tag{14}$$

For old classes, the previous prototype is moved into the current frame by

$$p_c = \mathrm{norm}(p_c^{t-1} A_t). \tag{15}$$

For new classes, p_c is the normalized centroid of current-backbone features. All class representatives now live in the same frame, but they are not used as classifier weights.

5.3 Radial Reliability

The transported prototype gives a class location, not its reliability. Under exact NC1, all classes are equally concentrated. In continual learning, replay and drift make this concentration class-dependent. We measure this local NC1 violation while keeping u_c as the global NC2/NC3 reference. Let $d_{\cos}(a,b) = 2 - 2\langle a, b\rangle$ denote the cosine-distance convention used by the radial head; for normalized vectors this is their squared Euclidean distance, and the implementation only clamps it at zero for numerical stability. Let $\mathcal{B}_c$ be the replay samples of class c available after task t. We estimate the raw class radius by

$$r_c = \frac{1}{|\mathcal{B}_c|} \sum_{(x_i, y_i) \in \mathcal{B}_c} d_{\cos}(z_i, p_c), \qquad z_i = \frac{f_\theta(x_i)}{\|f_\theta(x_i)\|}. \tag{16}$$

If a class is absent from replay, we use the global mean radius as a fallback.

The raw radii cannot be unconstrained class scales. Since the ETF allocates class directions uniformly, no class should win by claiming an arbitrarily large region of the sphere. We therefore convert raw radii into a *competitive* radius:

$$\rho_c = \bar{r} \frac{(r_c/\bar{r})^{\alpha_\rho}}{\frac{1}{|\mathcal{C}_t|}\sum_{j \in \mathcal{C}_t} (r_j/\bar{r})^{\alpha_\rho}}, \qquad \alpha_\rho = 0.25, \tag{17}$$

where $\bar{r}$ is the mean raw radius. This keeps the average radial budget fixed while allowing measured dispersion to modulate local support; the exponent $\alpha_\rho < 1$ makes the modulation conservative. Let $\bar{\rho} = \frac{1}{|\mathcal{C}_t|}\sum_{j \in \mathcal{C}_t} \rho_j$ denote the resulting mean competitive radius. This normalization is not a softmax: increasing one class radius increases its local support only relative to a fixed total radius budget, so the other classes must compensate through the denominator.

Finally, using p_c directly would let the local class measure replace the NC reference. As a stabilization step, we *retract* it toward u_c according to relative reliability:

$$\tilde{p}_c = \mathrm{norm}(\alpha_c p_c + (1 - \alpha_c) u_c), \qquad \alpha_c = \mathrm{clip}\left[\left(\frac{\bar{\rho}}{\rho_c + \varepsilon}\right)^{\gamma}, \alpha_{\mathrm{lo}}, \alpha_{\mathrm{hi}}\right], \tag{18}$$

with $\gamma = 0.5$ and $(\alpha_{\mathrm{lo}}, \alpha_{\mathrm{hi}}) = (0, 0.95)$. Classes with tight support keep more of their transported center, while high-radius classes are pulled back toward the symmetric ETF vertex. The upper clip prevents full trust in a drift-compensated prototype; the unconstrained variant is reported in the supplement. The radial score is the negative radius-normalized cosine distance to this NC-retracted center:

$$s_{\mathrm{rad}}(x,c) = -\frac{d_{\cos}(z, \tilde{p}_c)}{\rho_c + \varepsilon}. \tag{19}$$

A class scores highly only when the query is close to its NC-retracted prototype relative to its estimated radius. This term is class-local: it does not predict a task block or introduce a separate router.

5.4 Final Classifier

ETF scores and radial scores have different units, so we standardize each score vector per sample. This removes sample-specific scale and offset before composition; the final decision depends on the relative class ordering supplied by each evidence source, not on the arbitrary numerical units of cosine alignment versus radius-normalized distance:

$$\mathcal{N}[s](x,c) = \frac{s(x,c) - \frac{1}{|\mathcal{C}_t|}\sum_{j \in \mathcal{C}_t} s(x,j)}{\mathrm{Std}_{j \in \mathcal{C}_t}\, s(x,j) + \varepsilon}. \tag{20}$$

The final RCR score is

$$s_{\mathrm{RCR}}(x,c) = (1-\omega)\mathcal{N}[s_{\mathrm{ETF}}](x,c) + \omega \mathcal{N}[s_{\mathrm{rad}}](x,c), \qquad \omega = \frac{q}{q+1}, \tag{21}$$

where q is the number of classes per task. The scalar ω is fixed by task granularity and shared by all classes and samples. It is not selected from validation data: it increases the radial contribution when each task block contains more competing classes, and keeps more ETF weight in small-block settings such as Seq-CIFAR-10.

RCR does not retrain the backbone, fit a task oracle, use test-time task identity, or add a task-age prior. The only class-specific quantities are the ETF vertex u_c, transported prototype p_c, and reliability radius ρ_c. No scalar is selected per dataset: all reported datasets use the same fixed rules $\alpha_\rho = 0.25$, $\gamma = 0.5$, $(\alpha_{\mathrm{lo}}, \alpha_{\mathrm{hi}}) = (0, 0.95)$, and $\omega = q/(q+1)$.

NC consistency. RCR is designed to be inactive at the exact neural-collapse fixed point. Assume that transported class representatives coincide with ETF vertices, $p_c = u_c$, and that all classes have the same radius, $\rho_c = \rho$. Then $\tilde{p}_c = u_c$ by Equation (18) for any clipped value of α_c, and

$$s_{\mathrm{rad}}(x,c) = -\frac{2 - 2\langle z, u_c \rangle}{\rho} = \frac{2}{\rho} s_{\mathrm{ETF}}(x,c) - \frac{2}{\rho}. \tag{22}$$

The radial score is therefore a positive affine transform of the ETF score. Because Equation (21) uses per-sample standardization, $\mathcal{N}[s_{\mathrm{rad}}](x,\cdot) = \mathcal{N}[s_{\mathrm{ETF}}](x,\cdot)$, and RCR reduces exactly to the normalized nearest-ETF classifier. Thus, RCR changes decisions only when progressive neural collapse is imperfect, *i.e.* when transported class centers deviate from ETF vertices or class reliabilities become heterogeneous.

6 Experiments

6.1 Experimental Setup

Benchmarks and evaluation setting. We evaluate on Seq-CIFAR-10, Seq-CIFAR-100 [19], and Seq-TinyImageNet [20] with replay buffers of 200 and 500 samples. Seq-CIFAR-10 is split into 5 tasks with 2 classes per task, while Seq-CIFAR-100 and Seq-TinyImageNet use 10 tasks with 10 and 20 classes per task, respectively. To isolate classifier geometry, all paired head comparisons use the same learned progressive-ETF representations and differ only in the evidence used for global prediction. Reproduced artifacts follow the same ResNet-style backbone family as the reference implementation [14]. Additional implementation details, including the ProNC training objective and the values of λ_1, λ_2 used to produce the checkpoints, are given in the supplementary material (Section B).

Baselines and paired evaluation. For the literature tables, all non-Ours baselines are taken from [36], which reports the same datasets, memory budgets, and FAA/FF metrics. For the paired head comparison in Table 1, we reproduced the reference model with the official code over seeds $0,1,2$ and evaluated the ETF head and ProNC + RCR on the same artifacts. The reproduced ETF head is close to, but not identical to, the published row: across the six dataset–buffer settings, the absolute difference in final Class-IL FAA ranges from 0.29 to 2.45 points. We use published rows for literature positioning and reproduced artifacts for paired classifier comparisons.

Metrics. We report final average accuracy (FAA) and final forgetting (FF) in both Class-IL and Task-IL. Class-IL evaluates the global classifier over all seen classes. Task-IL restricts prediction to the ground-truth task block and therefore measures task-local class discrimination. To quantify the failure mode directly, we also report *route accuracy*: the fraction of test samples for which the globally predicted class belongs to the correct task block. A routing correction should increase route accuracy and Class-IL while leaving Task-IL nearly unchanged.

6.2 Main Results

Table 1: **RCR improves the global decision by repairing route accuracy.** On the same frozen ProNC checkpoints, Class-IL and Route increase together while Task-IL remains nearly unchanged. Parentheses report deltas against s_{ETF}; all entries are averaged over seeds $0,1,2$.

Dataset	Buffer	Score	Class-IL		Task-IL		Route
			FAA ↑	FF ↓	FAA ↑	FF ↓	↑
CIFAR-10	200	s_{ETF}	66.14	31.47	**96.74**	**0.62**	66.58
		s_{RCR} (Ours)	**70.36 (+4.22)**	**22.85 (-8.62)**	96.67 (-0.07)	0.73 (+0.11)	**70.82 (+4.24)**
	500	s_{ETF}	72.16	24.18	**96.82**	**0.47**	72.71
		s_{RCR} (Ours)	**73.68 (+1.52)**	**17.73 (-6.45)**	96.81 (-0.01)	0.54 (+0.07)	**74.33 (+1.62)**
CIFAR-100	200	s_{ETF}	41.71	38.55	85.53	4.51	42.81
		s_{RCR} (Ours)	**49.74 (+8.03)**	**16.27 (-22.28)**	**87.28 (+1.75)**	**2.36 (-2.15)**	**51.43 (+8.62)**
	500	s_{ETF}	48.65	23.16	85.54	4.29	50.14
		s_{RCR} (Ours)	**53.27 (+4.62)**	**10.18 (-12.98)**	**87.74 (+2.20)**	**2.02 (-2.27)**	**55.04 (+4.90)**
TinyImageNet	200	s_{ETF}	25.77	43.45	68.30	**9.58**	28.58
		s_{RCR} (Ours)	**31.42 (+5.65)**	**23.84 (-19.61)**	**69.59 (+1.29)**	**8.61 (-0.97)**	**34.76 (+6.18)**
	500	s_{ETF}	26.61	41.16	67.71	9.78	29.37
		s_{RCR} (Ours)	**32.24 (+5.63)**	**22.47 (-18.69)**	**70.16 (+2.45)**	**7.90 (-1.88)**	**35.46 (+6.09)**

6.2.1 Paired classifier comparison

Table 1 compares the reference ETF score and ProNC + RCR on identical evaluation artifacts. ProNC + RCR improves Class-IL FAA and Route together, while Task-IL changes little. This is the signature predicted by Equation (11): the gain comes from selecting the correct task block, not from relearning within-task boundaries.

Across datasets and memory budgets, Class-IL gains track Route gains. After progressive ETF training, the model often still separates classes within the correct task block, but the global score assigns samples to competing blocks. A compact visual summary and task-by-task diagnostic curves are provided in the supplementary material (Figures 1 and 2).

6.2.2 Comparison against the state of the art

The two tables below are for literature positioning. Their prior-method rows are copied from the published reference table, while the Ours row is computed on our three-seed reproduction. The perfectly paired comparison is therefore Table 1, where both heads are evaluated on the same reproduced checkpoints; the literature tables report whether the resulting classifier correction reaches or exceeds the published numbers under the same dataset and memory budgets.

Tables 2 and 3 report the main comparison for replay buffers of 200 and 500 samples. ProNC + RCR sets the best Class-IL FAA on five of six dataset-buffer settings and reduces Class-IL FF on all six relative to the reported reference. These tables should be read as the effect of a post-hoc, NC-consistent classifier correction applied to a trained progressive ETF model, not as a replacement training algorithm for continual learning. The largest gains appear on harder routing regimes, where many task blocks compete globally.

On Seq-CIFAR-100, Class-IL FAA improves by $+6.75$ points with buffer 200 and $+4.33$ with buffer 500; Class-IL FF is more than halved in both cases. Seq-TinyImageNet improves under both budgets, and Seq-CIFAR-10/200 gains $+4.78$ points. The only published-FAA exception is Seq-CIFAR-10/500, where the reported reference is 0.27 points higher. This is the benign regime predicted by the mechanism: only five task blocks compete, Task-IL is near 97%, and the Task-IL–Route gap is smallest. On matched checkpoints, RCR still improves this setting by $+1.52$ Class-IL FAA, $+1.62$ Route, and -6.45 Class-IL FF over the reproduced ETF head (Table 1).

6.3 Ablation

We ablate the correction on Seq-CIFAR-100 with buffer 200, the hardest CIFAR setting, while keeping the final checkpoint features fixed.

Table 4: **Each component targets routing.** Seq-CIFAR-100/200 ablation on the same checkpoint features.

Head	ETF	Transport	Radial	Comp. ρ_c	NC shrink	Class-IL ↑	Route ↑
ETF only	✓					41.71 ± 0.61	42.81 ± 0.57
Radial only		✓	✓	✓		46.21 ± 0.44	48.36 ± 0.49
No transport	✓		✓	✓		41.71 ± 0.61	42.81 ± 0.57
Equal radius	✓	✓	✓			44.49 ± 0.29	46.04 ± 0.28
No NC shrink	✓	✓	✓	✓		49.20 ± 0.48	50.93 ± 0.37
Ours (RCR)	✓	✓	✓	✓	✓	$\mathbf{49.74 \pm 0.58}$	$\mathbf{51.43 \pm 0.47}$

Note: Columns correspond to ETF evidence (Equation (13)), transport (Equations (14) and (15)), radial evidence (Equation (19)), competitive radii (Equation (17)), and NC shrinkage (Equation (18)).

Table 2: Literature comparison for buffer size 200. FAA denotes final average accuracy, and FF denotes final forgetting.

Setting	Method	Venue	Seq-CIFAR-10		Seq-CIFAR-100		Seq-TinyImageNet	
			FAA ↑	FF ↓	FAA ↑	FF ↓	FAA ↑	FF ↓
Class-IL	ER [29]	ICLR 2019	44.79 ± 1.86	59.30 ± 2.48	21.78 ± 0.48	75.06 ± 0.63	8.49 ± 0.16	76.53 ± 0.51
	iCaRL [28]	CVPR 2017	49.02 ± 3.20	23.52 ± 1.27	28.00 ± 0.91	47.20 ± 1.23	7.53 ± 0.79	31.06 ± 1.91
	DER [3]	NeurIPS 2020	61.93 ± 1.79	35.79 ± 2.59	31.23 ± 1.38	62.72 ± 2.69	11.87 ± 0.78	64.83 ± 1.48
	DER++ [3]	NeurIPS 2020	64.88 ± 1.17	32.59 ± 2.32	28.13 ± 0.51	60.99 ± 1.52	11.34 ± 1.17	73.47 ± 1.23
	Co^{2}L [1]	ICCV 2021	51.27 ± 1.86	30.17 ± 8.57	18.09 ± 0.49	64.14 ± 0.69	12.95 ± 0.06	62.04 ± 0.28
	CILA [57]	ICML 2024	59.68 ± 0.65	37.52 ± 5.84	19.49 ± 0.53	64.01 ± 0.18	12.98 ± 0.10	63.11 ± 0.61
	MNC^{3}L [6]	WACV 2025	52.20 ± 1.56	33.74 ± 1.65	15.81 ± 0.48	62.51 ± 0.40	10.57 ± 1.66	59.68 ± 2.02
	STAR [8]	ICLR 2025	62.10 ± 2.21	**21.78 ± 3.16**	18.29 ± 2.58	68.40 ± 8.52	11.55 ± 3.23	67.43 ± 1.48
	CSReL [53]	ICLR 2025	37.46 ± 1.57	26.34 ± 2.18	29.06 ± 1.00	58.23 ± 1.54	18.14 ± 3.10	49.77 ± 2.27
	NCT [40]	ICLR 2023	51.59 ± 0.41	22.48 ± 19.50	26.38 ± 0.57	27.40 ± 1.65	10.95 ± 1.45	49.33 ± 4.47
	ProNC [36]	ICLR 2026	65.58 ± 0.15	32.75 ± 4.71	42.99 ± 0.85	36.07 ± 0.51	27.44 ± 1.00	42.81 ± 0.56
	Ours	This work	**70.36 ± 0.62**	22.85 ± 1.45	**49.74 ± 0.71**	**16.27 ± 0.45**	**31.42 ± 0.80**	**23.84 ± 1.15**
Task-IL	ER [29]	ICLR 2019	91.19 ± 0.94	6.07 ± 1.09	60.19 ± 1.01	27.38 ± 1.46	38.17 ± 2.00	40.47 ± 1.54
	iCaRL [28]	CVPR 2017	88.99 ± 2.13	25.34 ± 1.64	51.43 ± 1.47	36.20 ± 1.85	28.19 ± 1.47	42.47 ± 2.47
	DER [3]	NeurIPS 2020	91.40 ± 0.92	6.08 ± 0.70	63.09 ± 1.09	25.98 ± 1.55	40.22 ± 0.67	40.43 ± 1.05
	DER++ [3]	NeurIPS 2020	91.92 ± 0.60	5.16 ± 0.21	66.80 ± 0.41	23.91 ± 0.55	43.06 ± 1.16	39.02 ± 0.43
	Co^{2}L [1]	ICCV 2021	84.69 ± 1.52	2.91 ± 5.25	49.19 ± 0.91	36.81 ± 0.63	37.07 ± 1.62	40.75 ± 3.55
	CILA [57]	ICML 2024	91.36 ± 0.08	5.49 ± 0.74	53.93 ± 1.02	33.07 ± 0.96	37.32 ± 1.87	41.40 ± 0.51
	MNC^{3}L [6]	WACV 2025	85.94 ± 0.22	4.90 ± 0.15	43.91 ± 0.76	39.79 ± 0.98	32.78 ± 3.54	45.10 ± 1.57
	STAR [8]	ICLR 2025	93.54 ± 1.48	6.23 ± 2.06	58.53 ± 13.06	27.84 ± 8.01	41.70 ± 0.95	34.78 ± 2.08
	CSReL [53]	ICLR 2025	69.22 ± 3.03	17.16 ± 2.91	66.99 ± 0.35	23.20 ± 1.74	45.04 ± 5.86	34.12 ± 2.18
	NCT [40]	ICLR 2023	80.63 ± 0.46	1.41 ± 1.09	75.75 ± 0.17	4.79 ± 0.07	52.71 ± 4.12	15.88 ± 0.95
	ProNC [36]	ICLR 2026	**96.86 ± 0.10**	**0.65 ± 0.08**	85.63 ± 0.73	4.40 ± 0.82	69.09 ± 0.65	9.42 ± 0.58
	Ours	This work	96.67 ± 0.09	0.73 ± 0.19	**87.28 ± 0.25**	**2.36 ± 0.14**	**69.59 ± 0.99**	**8.61 ± 1.09**

Table 3: Literature comparison for buffer size 500. FAA denotes final average accuracy, and FF denotes final forgetting.

Setting	Method	Venue	Seq-CIFAR-10		Seq-CIFAR-100		Seq-TinyImageNet	
			FAA ↑	FF ↓	FAA ↑	FF ↓	FAA ↑	FF ↓
Class-IL	ER [29]	ICLR 2019	57.74 ± 2.48	43.22 ± 2.10	22.35 ± 0.61	73.08 ± 0.78	9.99 ± 0.29	75.21 ± 0.54
	iCaRL [28]	CVPR 2017	47.55 ± 3.95	28.20 ± 2.41	33.25 ± 1.25	40.99 ± 1.02	9.38 ± 1.53	37.30 ± 1.42
	DER [3]	NeurIPS 2020	70.51 ± 1.67	24.02 ± 1.63	41.36 ± 1.76	49.07 ± 2.54	17.75 ± 1.14	59.95 ± 2.31
	DER++ [3]	NeurIPS 2020	72.70 ± 1.36	22.38 ± 4.41	38.20 ± 1.00	49.18 ± 2.19	19.38 ± 1.41	58.75 ± 1.93
	Co^{2}L [1]	ICCV 2021	61.78 ± 4.22	17.79 ± 3.36	26.64 ± 1.42	48.60 ± 1.34	18.71 ± 0.84	49.64 ± 1.14
	CILA [57]	ICML 2024	67.82 ± 0.33	18.22 ± 1.32	31.27 ± 0.17	45.67 ± 1.02	18.09 ± 0.49	64.14 ± 0.69
	MNC^{3}L [6]	WACV 2025	52.20 ± 1.56	27.88 ± 2.75	22.29 ± 0.18	46.09 ± 0.58	11.52 ± 0.01	44.96 ± 0.09
	STAR [8]	ICLR 2025	69.15 ± 3.53	18.59 ± 2.91	28.45 ± 1.70	53.11 ± 1.57	15.19 ± 2.61	63.42 ± 3.19
	CSReL [53]	ICLR 2025	40.82 ± 4.09	29.18 ± 6.91	39.22 ± 1.70	47.03 ± 2.57	22.13 ± 0.35	50.01 ± 1.87
	NCT [40]	ICLR 2023	60.93 ± 0.94	**13.82 ± 3.64**	33.84 ± 0.38	24.37 ± 5.42	18.24 ± 0.62	50.90 ± 1.16
	ProNC [36]	ICLR 2026	**73.95 ± 0.68**	26.75 ± 1.29	48.94 ± 0.44	24.32 ± 0.24	29.06 ± 0.32	38.58 ± 0.39
	Ours	This work	73.68 ± 0.88	17.73 ± 0.88	**53.27 ± 0.41**	**10.18 ± 0.38**	**32.24 ± 0.05**	**22.47 ± 0.13**
Task-IL	ER [29]	ICLR 2019	93.61 ± 0.27	3.50 ± 0.53	73.98 ± 1.52	16.23 ± 1.06	48.64 ± 0.46	30.73 ± 0.62
	iCaRL [28]	CVPR 2017	88.22 ± 2.62	22.61 ± 3.97	58.16 ± 1.76	27.90 ± 1.37	31.55 ± 3.27	39.44 ± 0.84
	DER [3]	NeurIPS 2020	93.40 ± 0.21	3.72 ± 0.55	71.73 ± 0.74	25.98 ± 1.55	51.78 ± 0.88	28.21 ± 0.97
	DER++ [3]	NeurIPS 2020	93.88 ± 0.50	4.66 ± 1.15	74.77 ± 0.31	15.75 ± 0.48	51.91 ± 0.68	25.47 ± 1.03
	Co^{2}L [1]	ICCV 2021	89.51 ± 2.45	2.65 ± 1.00	62.32 ± 0.19	23.47 ± 0.27	50.74 ± 1.24	13.80 ± 2.10
	CILA [57]	ICML 2024	93.29 ± 0.24	0.65 ± 0.13	68.29 ± 0.46	16.98 ± 0.56	49.19 ± 0.91	27.83 ± 1.32
	MNC^{3}L [6]	WACV 2025	85.94 ± 0.22	3.15 ± 0.42	56.43 ± 0.29	24.29 ± 0.50	36.32 ± 0.05	39.00 ± 0.17
	STAR [8]	ICLR 2025	95.36 ± 0.37	2.38 ± 0.34	74.06 ± 1.63	15.22 ± 0.96	55.06 ± 2.07	15.19 ± 2.61
	CSReL [53]	ICLR 2025	73.30 ± 5.92	16.89 ± 3.19	75.09 ± 0.78	15.55 ± 0.76	51.94 ± 0.22	28.20 ± 0.36
	NCT [40]	ICLR 2023	81.27 ± 0.24	3.84 ± 0.35	76.06 ± 0.52	3.97 ± 0.75	62.30 ± 0.41	**6.84 ± 0.87**
	ProNC [36]	ICLR 2026	**96.95 ± 0.14**	**0.24 ± 0.14**	86.38 ± 0.43	4.35 ± 0.17	69.77 ± 0.89	9.52 ± 0.96
	Ours	This work	96.81 ± 0.04	0.54 ± 0.08	**87.74 ± 0.07**	**2.02 ± 0.16**	**70.16 ± 0.20**	7.90 ± 0.16

Table 4 isolates the mechanism. Transported radial evidence gives the largest gain over ETF-only scoring. A shared radius removes class-dependent reliability and degrades Route. Removing NC retraction causes a smaller but consistent drop. We therefore interpret the retraction as an NC-consistent stabilization term rather than as the primary source of the gain: the main improvement comes from transport, radial reliability, and the competitive radius. Radial evidence without transport is inert in the composed head: its center is the ETF vertex, so after score normalization it matches the ETF decision.

We also test the simplest alternative explanation: a replay-fitted task bias has direct access to replay labels, yet worsens held-out Route (Table 5). This matches Figure 4: replay samples are too clean to expose test-time routing drift.

Table 5: **Replay-fitted calibration does not explain RCR.** A task-bias calibration fit on replay labels is evaluated on held-out Seq-CIFAR-100/200 test features at the final task. It worsens routing, confirming that the replay buffer is not a reliable calibration set.

Head	Class-IL ↑	Task-IL ↑	Route ↑	Wrong-task ↓
ProNC ETF	41.71 ± 0.61	85.53 ± 0.16	42.81 ± 0.57	57.19 ± 0.57
Replay task-bias	29.91 ± 0.64	85.53 ± 0.16	30.66 ± 0.59	69.34 ± 0.59
Ours (RCR)	$\mathbf{49.74 \pm 0.58}$	$\mathbf{87.28 \pm 0.20}$	$\mathbf{51.43 \pm 0.47}$	$\mathbf{48.57 \pm 0.47}$

Full classifier-variant benchmarks and sensitivity studies are in the supplement. We additionally include a small NCT diagnostic to check that the route-correction signal is not tied to one progressive-ETF implementation. On Seq-CIFAR-100 with buffer 200, an offline RCR-style ramp applied to a reproduced NCT-style progressive ETF model improves Class-IL FAA by +2.40, reduces Class-IL FF by 1.21, and improves route accuracy by +2.32 (Table 4). This is a scope diagnostic, not a full benchmark of NCT.

6.4 Mechanistic Analysis

The mechanism has two parts. The ETF score supplies a globally comparable NC frame, while the transported radial score asks whether z is locally plausible under the current class measure. Score normalization makes the two evidences commensurate before composition. In the exact NC limit, the radial score is a positive affine transform of the ETF score, so RCR changes no decisions.

Table 6: **RCR is most active when routing drift is largest.** In the exact NC limit, transported class representatives coincide with ETF vertices and RCR reduces to nearest-ETF inference. Across validated runs, the head is least active when the ProNC task-local classifier already routes well, and its gains grow with the held-out Task-IL–Route gap (row-wise $r = 0.97$ with ΔRoute and $r = 0.96$ with ΔClass). Values are final-task mean ± std over three seeds.

Dataset	Buf.	$\bar{\delta}$	NC1 disp.	CV(ρ)	Task–Route gap	ΔRoute	ΔClass
Seq-CIFAR-10	200	1.50 ± 0.01	0.37 ± 0.02	0.13 ± 0.00	30.16 ± 1.24	$+4.23 \pm 1.41$	$+4.22 \pm 1.40$
Seq-CIFAR-10	500	1.44 ± 0.01	0.38 ± 0.02	0.14 ± 0.01	24.11 ± 1.36	$+1.62 \pm 1.14$	$+1.52 \pm 1.05$
Seq-CIFAR-100	200	1.61 ± 0.00	0.41 ± 0.00	0.11 ± 0.00	42.71 ± 0.41	$+8.62 \pm 0.11$	$+8.03 \pm 0.19$
Seq-CIFAR-100	500	1.52 ± 0.00	0.43 ± 0.00	0.10 ± 0.00	35.41 ± 0.47	$+4.90 \pm 0.32$	$+4.62 \pm 0.40$
Seq-TinyImageNet	200	1.63 ± 0.00	0.43 ± 0.00	0.08 ± 0.00	39.72 ± 0.65	$+6.18 \pm 0.72$	$+5.65 \pm 0.69$
Seq-TinyImageNet	500	1.60 ± 0.00	0.43 ± 0.00	0.07 ± 0.00	38.33 ± 0.73	$+6.09 \pm 0.36$	$+5.63 \pm 0.35$

Note: $\bar{\delta}$ is the mean cosine distance $d_{\cos}(p_c, u_c) = 2 - 2\langle p_c, u_c \rangle$ between transported class representatives and ETF vertices. Task–Route gap is ProNC Task-IL FAA minus ProNC route accuracy.

Table 6 makes the link quantitative across datasets: the smallest gains occur when the ProNC task-local classifier already routes well, and gains grow with the Task-IL–Route gap. Detailed stage-level NC diagnostics, class-wise correlations, and true-route headroom are reported in the supplementary material (Tables 11 and 12).

Seq-CIFAR-10 is closest to the benign regime: only five task blocks compete, Task-IL is near 97%, and route accuracy is much higher than on Seq-CIFAR-100 or Seq-TinyImageNet. Accordingly, RCR gives smaller gains there, especially with buffer 500. The gains grow when more task blocks must be compared and the Task-IL–Route gap is larger. This supports the intended interpretation: RCR is an NC-deviation correction, not a new within-task classifier. When ETF geometry and class reliabilities are already compatible, it stays nearly inactive; when transported centers and radii depart from the NC ideal, it changes the global route. The true-route bounds in the supplement show that substantial routing headroom remains after RCR, so the post-hoc correction does not saturate the problem. The supplementary class-wise correlations also show that center drift, radius growth, and route error increase with class age; RCR targets this old-class reliability decay without using task age as an explicit score bias.

7 Discussion

RCR is intentionally scoped to neural-collapse continual learning. It is not a universal calibration layer for arbitrary continual learners: it uses the explicit ETF geometry, transported class measures, and replay-estimated radii of a progressive NC model. This scope is deliberate. It isolates the scoring failure from representation learning and shows that routing drift can be partly repaired at the classifier level when the backbone already preserves task-local discrimination.

In this controlled setting, RCR uses the same replay-derived artifacts as the evaluated progressive NC model to estimate class radii and linear feature transport; inference then adds only one extra class-prototype score matrix, per-sample standardization, and no extra backbone forward. The stored class-level state is small: for TinyImageNet with $C = 200$ and $d = 512$, one transported representative plus one radius per class is about 0.39 MB in FP32. The route-oracle analysis in the supplement should be read constructively: a gap remains to the true-route upper bound, so a single post-hoc score does not close routing drift. Instead, classifier-level reliability recovers a measurable fraction of the lost route while preserving Task-IL, leaving room for training-time objectives that make the representation itself more route-stable.

8 Conclusion

We identified routing drift as a measurable failure mode of neural-collapse continual learning: Task-IL can remain high while global Class-IL fails because samples route to the wrong task block. We proposed Reliability-Calibrated Routing, which preserves the global ETF frame and complements it with transported radial class plausibility. Across Seq-CIFAR-10, Seq-CIFAR-100, and Seq-TinyImageNet replay settings, this post-hoc NC-consistent correction gives the best Class-IL FAA on five of six benchmark settings and consistently reduces Class-IL forgetting without sacrificing task-local discrimination. A natural next step is to turn the same route-reliability signal into a training-time objective, so the representation be-comes more route-stable rather than only corrected at inference. It will also be important to test whether this reliability-calibrated view extends to other neural-collapse classifiers, larger visual streams, and settings where class measures must be estimated with weaker replay support.

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